



Monitor and Control (MC)

MC Overview

Required PA Information

Intent

Provides an understanding of the project progress so appropriate corrective actions can be taken when performance deviates significantly from plans.

Value

Increases the probability of meeting objectives by taking early actions to adjust for significant performance deviations.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

- MC 1.1 Record task completions.
- MC 1.2 Identify and resolve issues.

Level 2

- MC 2.1 Track actual results against estimates for size, effort, schedule, resources, knowledge and skills, and budget.
- MC 2.2 Track the involvement of identified stakeholders and commitments.
- MC 2.3 Monitor the transition to operations and support.
- MC 2.4 Take corrective actions when actual results differ significantly from planned results and manage to closure.

Level 3

- MC 3.1 Manage the project using the project plan and the project process.
- MC 3.2 Manage critical dependencies and activities.
- MC 3.3 Monitor the work environment to identify issues.
- MC 3.4 Manage and resolve issues with affected stakeholders.

Additional PA Explanatory Information

The term “project plan” refers to the overall plan for managing the project and includes a coherent description of who does what and when for developing, updating, or delivering a solution. The project plan can be a stand-alone document or distributed across multiple documents and provides a way to track and communicate actual progress and determine if corrections are needed.

Tracking progress typically consists of comparing actual values to planned or estimated values for:

- Size
- Complexity
- Effort
- Cost
- Schedule
- Quality
- Milestones
- Knowledge and skills
- Resources
- Stakeholder involvement
- Commitments
- Transition to operations and support

Tracking actuals against estimates supports managing the expectations of customers and stakeholders. Take corrective actions when actual values deviate significantly from expected values. These corrective actions may include:

- Revising the strategy for accomplishing the work
- Updating objectives
- Revising the plan
- Revising the estimates
- Establishing or modifying agreements and commitments
- Updating risk and opportunity management activities and work products

If corrective actions are required to resolve significant deviations from project plans, define and track these actions to closure.

Related Practice Areas

Managing Performance and Measurement (MPM)

Planning (PLAN)

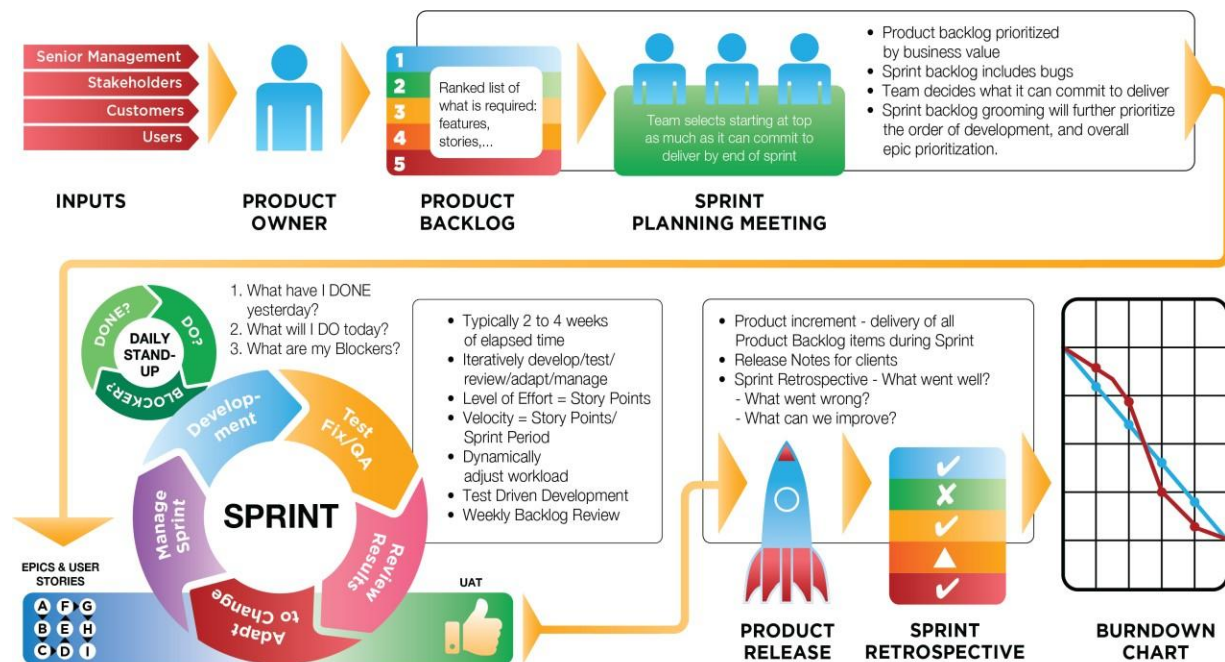
Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

In [Figure MC-1: Monitoring Agile Development](#), each of the activities in the development flow provide opportunities for monitoring progress and are reviewed during the retrospective. For example, during the daily stand-up, individuals report on progress and blockers, providing overall status information. During the Sprint retrospective, the team reviews progress made from the last Sprint and adjusts future Sprints and epics to account for actual versus planned velocity.

Figure MC-1: Monitoring Agile Development

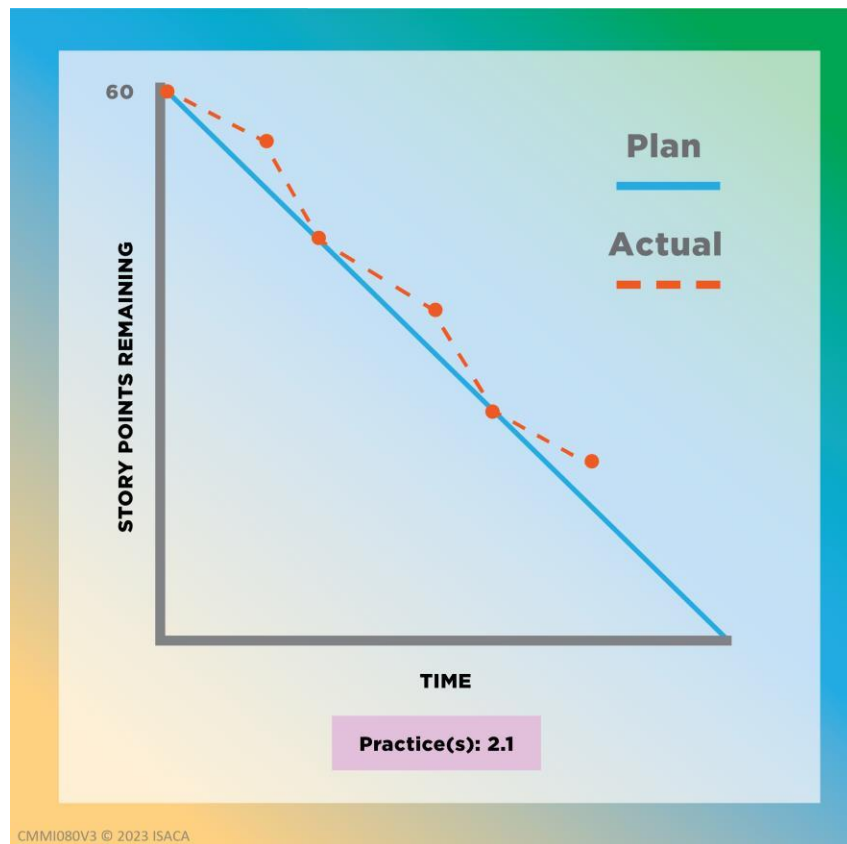


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Typical monitoring practices for agile teams, e.g., stand-up meetings, visual information management, is reported using:

- A burndown chart (refer to [Figure MC-2: Burndown Chart](#)) showing the number of story points remaining, tracked within each Sprint, and representing all of the work for a release typically consisting of several Sprints. A burndown chart is updated daily indicating the time needed to complete the work committed for the Sprint.
- Visual information in Kanban boards or other tracking tools indicate the current state of team performance, and related factors that can impact performance or progress

Figure MC-2: Burndown Chart

**Data**

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

The production of approaches for managing data to meet relevant needs of the organization is not a once-and-done activity. Once developed, the execution of the approach and the changing needs of the business must be closely monitored and regularly evaluated to ensure the approach meets the business needs for quality. As shortfalls are identified or gaps are discovered, the approaches must evolve, which may involve changes of focus or adjustments to ongoing projects.

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams use automated tools to monitor their DevSecOps pipelines and collect and assess key information about application use to discover trends and identify problem areas. DevSecOps teams monitor their infrastructure resources, network transport, applications and microservices, containers, interfaces or connections, gate-check points, endpoint behavior, and security event logs to take actions when needed. This approach allows teams to “shift left” to earlier stages in development and minimize broken production changes.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Develop monitoring and control functions early in the project during planning when defining the supplier management strategy.

Monitor and control activities are essential throughout the supplier management process to ensure:

- Application of appropriate resources
- Acquirer activities progress according to plan

After selecting one or more suppliers and establishing agreements, the acquirer continues to monitor and control its activities and work products. At the same time, the acquirer monitors and controls supplier progress and performance for effects to the overall effort.

Define supplier progress and performance reporting requirements in the supplier agreement consistent with the needs of the contract.

Level 1

MC 1.1

Required Practice Information

Practice Statement

Record task completions.

Value

Enables the team and senior management to make better decisions to achieve objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Tracking completed work is part of monitoring progress. Regularly review tasks to determine status, which may include:

- Completed
- Delayed
- Not completed

Example Activities

Example Activities	Further Explanation
Record task completion.	Although tasks can be given a percentage completion, this may lead to inaccurate status reporting such as "90% done 90% of the time." One way of avoiding this is to only show 100% complete or not complete.
Review updated task list with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Task list	Includes: • Description • Status • Date

MC 1.2

Required Practice Information

Practice Statement

Identify and resolve issues.

Value

Supports prevention of uncontrolled cost and schedule creep.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify and resolve issues as they arise. Resolving issues is important to keeping tasks on schedule. Analyze issues as they are identified to determine the appropriate corrective action and track to closure.

Issues may generate unplanned work. If issues are not monitored and controlled, work can be delayed without an understanding of the reason.

Example Activities

Example Activities	Further Explanation
Record the issue in the issue and action item list.	
Assign responsibility for resolving the issue or action item.	Ensure people are aware that they have been given responsibility to resolve the action.
Assign a due date.	Work with the person assigned responsibility to determine when the issue or action can be completed.
Track issues and action items to closure.	Tracking to closure is critical to knowing if and how the project plan will be impacted.

Example Work Products

Example Work Products	Further Explanation
Issues and action item list	Includes the: • Description of the issue or action item • People assigned to the issue or action item • Due date • Status

Level 2

MC 2.1

Required Practice Information

Practice Statement

Track actual results against estimates for size, effort, schedule, resources, knowledge and skills, and budget.

Value

Identifies significant deviations so more effective corrective actions can be taken which increases the likelihood of meeting objectives.

Additional Required Information

Indicators of project progress and performance include characteristics of work products and tasks that may include:

- Cost
- Budget
- Effort
- Schedule
- Size
- Complexity
- Capacity and availability
- Function
- Knowledge and skills
- Resources
- Stakeholder involvement
- Commitments
- Transition to operations and support

Explanatory Practice Information

Additional Explanatory Information

Record associated contextual information when tracking actual results versus estimates to help understand what the measurement data represents.

Frequency of monitoring depends on the:

- Collection schedule
- Pace and duration of the work
- Agreed to milestones

Example Activities

Example Activities	Further Explanation
Track actual results to plans and estimates.	Track size, effort, schedule, resources, knowledge and skills, and budget. Tracking actual results typically occur through: • Progress reporting • Status reviews • Milestone reviews
Monitor resource capacity, and availability.	Identify what resources are needed to address the corrective actions. This typically includes: • People • Processes • Physical facilities • Computers, peripherals, and software • Networks • Security environment
Monitor the knowledge and skills of workgroup members.	This typically includes: • Periodically measuring the knowledge and skills of people to assess changes • Comparing actual training obtained to training recorded in the project plan • Identifying significant differences from project plan estimates
Monitor commitments against those identified in the project plan.	This includes: • Regularly reviewing internal and external commitments • Identifying commitments that have not been satisfied or are at significant risk of not being satisfied • Monitoring availability, reliability, and maintainability against their requirements • Recording review results
Record significant differences in planned vs. actual values.	This includes: • Defining criteria for what "significant" means for the planned and actual values • Keeping a record of significant differences to be used for more effective future planning
Monitor progress against schedule.	This typically includes: • Periodically measuring the actual completion of activities and milestones • Comparing actual completion of activities and milestones to the project schedule to identify significant deviations
Monitor expended effort and costs.	This typically includes:

Example Activities	Further Explanation
	- Periodically measuring the actual effort and costs expended - Comparing actual effort and costs, to the planned/estimated budget and costs - Identifying significant deviations from the project budget and estimates

Example Work Products

Example Work Products	Further Explanation
Records of actuals versus estimates	Typically includes the: - Budget - Schedule - Size - Effort - Resources - Knowledge and skills
Records of significant deviations	
Records of status reviews	
Corrective actions	
Cost performance reports	Include planned vs. actual results on cost of tasks, activities, and deliverable dates, their sequence, and resources needed.
Schedule performance reports	Include planned vs. actual results on schedule of tasks, activities, and deliverable dates, their sequence, and resources needed.

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Monitoring and analyzing capacity and availability can help identify the need for corrective actions to prevent service interruption and service system failure.

Record the use of each resource including the use of each resource by each component, e.g., the extent or degree of use by each component for a given resource. Analyze the effect of failures to align capacity and availability.

Monitor the use of resources during unexpected increases in demand to determine whether corrective actions are needed. Examples of corrective actions include:

- Adjustments to resources provided
- Adjustments to thresholds

- Adjustments to descriptions of the normal use of service resources and service system performance

Identify the need for corrective actions:

- Based on monitoring and analyzing capacity and availability
- In response to service incidents, change requests, changes to service requirements (current and future)
- To improve service system performance or prevent breaches of the service agreement

Monitor the service system to detect or prevent the failure of components that affect service system availability. At a minimum, monitor availability. Monitor other quality attributes if appropriate based on the type of service, development, or acquisition provided. For many service systems, it may also be appropriate to monitor other quality attributes such as reliability and maintainability. Monitor the resilience of the service system to service component failure and identify the impacts of specific failures on service system availability.

Activities to monitor capacity and availability of service systems may include:

- Monitoring the use of resources against thresholds, descriptions of normal use, and service system performance
- Estimating future changes, either growth or reduction, in resource use
 - Methods and tools for estimating service system behavior include trend analysis, analytical modeling, simulation modeling, baseline models, and application sizing
 - Resource usage growth estimates can be based on collected capacity and availability data, projected requirements, and service system representations
- Communicating the analysis of results on performance objectives and their impact on capacity and availability
 - Capacity and availability reports can be regular or ad hoc. If helpful, simplify reporting by using databases with automated reporting features. Follow organizational reporting standards. When they exist, use standard tools and techniques to integrate and consolidate information in the reports.
 - Information should be appropriate to the audience and understandable, and it may need to address multiple perspectives. These perspectives can include business, end user, customer, or provider. Agreements can define the reported information, to whom it should be delivered, and how it is provided, e.g., format, detail, distribution, media.
 - Availability is typically reported as a percentage. If required, in addition to reporting availability, report on reliability, e.g., reliability of the service or service system components, maintainability, and other quality attributes.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Track resource commitments that result in expenditures, e.g., issued purchase orders and completed, accepted supplier deliverables, when the organization incurs the expense. Track resource commitments even before formal payment, to account for financial and legal obligations. Monitor commitments that do not result in expenditures, e.g., allocation of resources or skill sets.

Example supplier deliverables include:

- Supplier progress and performance reports
- Records of significant deviations from plans or processes
- Cost performance reports

MC 2.2

Required Practice Information

Practice Statement

Track the involvement of identified stakeholders and commitments.

Value

Manages stakeholder involvement critical to successful work completion.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Manage stakeholder involvement according to the plan and process. Re-plan stakeholder involvement when changes to requirements, situation, or status occur.

Example Activities

Example Activities	Further Explanation
Periodically review and record the status of stakeholder involvement.	Stakeholder involvement can be tracked in events such as team meetings and cross-functional coordination meetings.
Identify and record significant stakeholder issues.	
Develop recommendations and coordinate actions to resolve issues.	

Example Work Products

Example Work Products	Further Explanation
Records of stakeholder involvement	Includes meeting records and reviews with attendee lists.
Agendas and schedules for collaborative activities	
Recommendations for resolving stakeholder issues	Includes records of decisions.
Recorded issues	Identifies both resolved and unresolved issues.

Related Practice Areas

Estimating (EST)

Planning (PLAN)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Monitoring and tracking stakeholder involvement involves communicating critical information to stakeholders. Communicating the results of data on capacity and availability management helps focus on meeting performance objectives.

Example stakeholder communication activities and topics include:

- Reporting the performance and use of resources
- Reporting exception conditions in the service system and requirement breaches
- Reporting data from monitoring against growth estimates in resource use
- Reporting the capacity, availability, reliability, and maintainability of resources, including:
 - Performance reports
 - Resource use reports
 - Resource use projections
 - Availability reports

MC 2.3

Required Practice Information

Practice Statement

Monitor the transition to operations and support.

Value

Ensures expected benefits are obtained by smooth solution transitions and successful implementations.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Monitor and control the actual transition of the accepted solution against the plan for the transition to operations and support. In some instances, this may be accomplished by delivering the solution directly to the customer.

Example Activities

Example Activities	Further Explanation
Monitor the capabilities of operations and support to receive, store, use, and keep updated new or modified solutions.	• Operations and support organizations should demonstrate readiness to accept the solution and provide uninterrupted support • Use transition readiness criteria and verification and validation practices to decide if delivered solutions meet specified requirements • Use verification and validation practices to confirm readiness for acceptance into operations and support
Monitor training delivery to stakeholders involved in receiving, storing, using, and updating solutions.	This typically includes: • Verifying correct training materials and resources specific to stakeholders involved are available and used • Verifying that the right training is given to the right people at the right time • Verifying that the delivered training has enabled the recipients to carry out their work efficiently and effectively
Review and analyze the results of transition activities.	Decide whether corrective actions must be completed before transferring responsibility to operations and support. Example work products that support transition analyses include: • Transition activity reports that include quality measures collected during the pilot and the warranty period • Problem tracking reports, detailing resolution time, escalation, and cause analysis • Change management reports • Configuration management records • Operation logs to decide that sufficient information is stored to support revisions • Security reports • Actual operations and support costs compared to estimates

Example Work Products

Example Work Products	Further Explanation
Status reports of transition activities	Current status of transition activities, including any risks, issues, or corrective actions taken.
Transition readiness report	Description of the state of readiness of the solutions prior to transition to ensure it will happen in accordance with the plan.
Records of transition support reviews	May include corrective actions.
Lessons learned report	

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Typically, the supplier has a role in integrating and packaging solutions and preparing for the transition to operations and support, including support for business user acceptance. The acquirer monitors these supplier activities. The supplier request package and the supplier agreement set the expectations of the supplier and the acceptance criteria to transition to operations and support.

Example supplier deliverables include:

- Training materials and supporting work products
- Site readiness reports
- Verification reports
- Training records
- Operational readiness reports
- Test results
- Pilot results

The acquirer makes adequate provisions to operate the acquired solution through the supplier agreement or in-house operations and support organizations. Typically, the supplier develops training for the solution.

MC 2.4

Required Practice Information

Practice Statement

Take corrective actions when actual results differ significantly from planned results and manage to closure.

Value

Manages corrective actions to increase the probability that objectives will be met.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Collect and analyze issues and decide on corrective actions to address them. Manage the corrective actions to closure. Actions can be automated, performed manually, or a combination of the two. Examples of corrective actions may include:

- Adjustments to resources to prevent performance problems or improve performance
- Rebalancing workload among resources
- Improving processes to allow for greater productivity, efficiency, and effectiveness
- Improving design such as making use of new technologies to allow for greater productivity, efficiency, or effectiveness
- Addressing capacity, and adding availability, e.g., adding people, or other resources
- Tuning to optimize and improve capacity or performance
- Adjusting requirements
- Improving the use of resources through demand management techniques

Example Activities

Example Activities	Further Explanation
Collect issues for analysis.	Collect issues from reviews and the execution of other processes. Examples of issues to collect include: • Issues discovered when performing technical reviews, verification, and validation • Significant deviations in project planning parameters from estimates in the project plan • Commitments (either internal or external) that have not been satisfied • Significant changes in risk status • Data access, collection, privacy, or security issues • Stakeholder representation or involvement issues • Solution, tool, or environment transition assumptions (or other customer or supplier commitments) that have not been achieved
Analyze issues to decide if corrective action is needed.	Corrective action is required when the issue, if left unresolved, may prevent the project from meeting its objectives.

Example Activities	Further Explanation
Take corrective action on identified issues.	Steps typically include: • Deciding and recording actions to resolve selected issues. Examples of potential actions include: ○ Modifying the Statement of Work (SOW) ○ Modifying requirements ○ Revising estimates and plans ○ Renegotiating commitments ○ Adding resources ○ Changing processes ○ Improving skills and efficiency ○ Revising project risks • Obtaining agreement from affected stakeholders on the actions • Using the organization's defined and established methods to resolve conflicts and disputes • Negotiating changes to internal and external commitments
Manage corrective actions to closure.	This typically includes: • Tracking corrective actions to completion • Analyzing results of corrective actions to determine their effectiveness and if additional corrective actions are needed • Recording final resolution of significant deviations when initial corrective action taken was not effective

Example Work Products

Example Work Products	Further Explanation
List of issues requiring corrective action	Include the: • Status of issue or action item • Person responsible for the action item • Corrective action plans • Corrective action results

Related Practice Areas

Planning (PLAN)

Verification and Validation (VV)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Work group level monitoring and control or measurement and analysis can adequately cover some monitoring of service system operation. This can include managing and controlling other operationally-oriented quality attributes associated with service delivery, such as:

- Capacity
- Availability
- Responsiveness
- Service Level Agreement performance
- Usability
- Reliability
- Maintainability
- Safety
- Security
 - Monitoring for security breaches, correcting vulnerabilities, and controlling access to services
 - Ensuring that the service provider only delivers approved services, as specified in the service agreement, to authorized personnel

However, some services can require monitoring and data collection at the level of individual service requests or continuously within the scope of a single service request. Such monitoring can require its own tools to handle data collection, analysis, and reporting appropriately. These tools are often automated. Perform low-level monitoring of service system components using monitoring and data collection tools as appropriate.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer has the sole responsibility for taking corrective actions when either the acquirer or supplier implementation results deviate from plan.

The acquirer determines, e.g., by monitoring measurement data, whether supplier progress is sufficient to meet a service level defined in the supplier agreement. If the supplier's progress is not sufficient, the acquirer initiates and manages corrective action with the supplier. If the supplier does not comply appropriately with this corrective action, the acquirer escalates the issue for resolution.

Level 3

MC 3.1

Required Practice Information

Practice Statement

Manage the project using the project plan and the project process.

Value

Ensures necessary activities are performed which reduces rework and improves the likelihood of achieving objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Managing the project helps to:

- Understand how much time is spent on each activity and decide if it is the most effective use of time
- Measure the resources used and available to a project
- Provide status to members of the team and stakeholders

Example Activities

Example Activities	Further Explanation
Manage the project activities using the project process and all related plans.	The project process is tailored by the project from the organization's set of standard processes. This includes: • Using the defined entry and exit criteria to start and finish tasks • Monitoring activities that could significantly affect actual values of the planning parameters, such as size, effort remaining, effort expended and changes in complexity • Tracking planning parameters using measurable thresholds to trigger investigation and corrective action • Monitoring risks • Managing external and internal commitments based on plans for tasks and work products • Understanding the relationships between the: ○ Tasks and work products of the project process, and

Example Activities	Further Explanation
	○ Roles of affected stakeholders ● Using quality control processes, e.g., peer reviews, objective evaluations, to improve both visibility into performance and control of the work ● Taking corrective action when progress deviates significantly from the plan
Collect and analyze selected measurements to manage the project and support the organization's needs.	
Periodically review and align project performance with organizational, customer, and end user requirements and objectives.	Examples of actions that achieve alignment include: ● Updating the schedule in response to changes in planning parameters, project, and risks ● Updating requirements or commitments in response to changes in market opportunities or customer and end user needs ● Terminating the project, iteration, or release
Resolve causes of issues affecting project objectives.	This includes: ● Reviewing issues and lessons learned from previous work ● Performing causal analysis of selected issues to identify where corrective actions need to be taken ● Evaluating process changes needed to prevent recurrence ● Taking the corrective actions and implement process changes ● Ensuring that the corrective actions and process changes have prevented recurrence of issues and improved performance

Example Work Products

Example Work Products	Further Explanation
Results of monitoring	
Collected measures and status records or reports	

Related Practice Areas

Causal Analysis and Resolution (CAR)

Managing Performance and Measurement (MPM)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

When managing the project, consider the critical dependencies of service delivery and service performance, such as:

- Timing of shipments
- Service delivery activities
- Service delivery schedules
- Operating procedures
- Service requests identified in service agreements
- Service delivery performance and measurements
- Locations of facilities for service delivery

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Example supplier deliverables include:

- Project progress and performance reports
- Measures
- Risk logs
- Issue logs
- Decision logs

MC 3.2

Required Practice Information

Practice Statement

Manage critical dependencies and activities.

Value

Manages critical dependencies to significantly reduce risk and increase the likelihood of meeting objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Critical dependencies can also involve on-time availability of resources.

Example Activities

Example Activities	Further Explanation
Review and update dependencies.	Record agreements for: • Coordinating dependent work • Ensuring that resources are available on time Provide advance communication to all affected stakeholders when dependencies cannot be met.
Record minutes from reviews and discussions.	
Record issues.	

Example Work Products

Example Work Products	Further Explanation
Updated critical dependencies	
Recorded agendas and meeting minutes	Dependencies are commonly managed and coordinated in: • Status reviews • Management reviews • Affected stakeholder discussions • Cross-functional team coordination events
Recorded issues	This may include: • Supplier delays • Stakeholder involvement (or lack thereof) • Recommendations for resolving issues

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The supplier agreement provides the basis for managing supplier involvement in the project. Supplier agreements, e.g., interagency and intercompany agreements, memoranda of understanding, memoranda of agreement, that the acquirer makes with stakeholder organizations provide the basis for stakeholder organization involvement. These stakeholder organizations can be solution providers or recipients. These agreements are particularly important when the acquirer's project produces complex integrated solutions.

MC 3.3

Required Practice Information

Practice Statement

Monitor the work environment to identify issues.

Value

Ensures objectives are met by providing an effective, safe, and healthy work environment.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Identify and correct environmental and physical factors that degrade the safety, health, effectiveness, and productivity of the workforce.

Monitor the environment to ensure that people can focus on achieving objectives and are free from issues and unwanted distractions. Ensure the entire workforce recognizes that worker safety and health is central to objective accomplishment.

Example Activities

Example Activities	Further Explanation
Monitor work environment elements that affect safety, health, effectiveness, and productivity and identify and record any corrections needed.	Work environment elements may include: • Standard workstation hardware and software • Standard production and calibration equipment • Buildings, facilities, and other physical resources • Security • Safety • Management and leadership • Specific environmental conditions • Periodic inspections of the work environment • Health and welfare • Diversity • Privacy
Monitor physical factors in the work environment that could degrade performance and identify and record any corrections needed.	Examples of physical factors that could degrade performance include: • Inadequate office or meeting space • Poor lighting • Inadequate heating, ventilation, or cooling • Unpleasant odors or fumes • Vibration • Excessive noise • Crowding or isolation • Environmental hazards

Example Activities	Further Explanation
Identify, record, and report potential work environment issues and corrections needed.	Examples of potential work environment issues problems include: • Failure to apply required safety standards • Inadequate security • Improper ergonomics • Exposure to unhealthy substances • Poor air or water quality • Excessive stress
Take reasonable steps to accommodate work environment issues while corrections are being made.	
Remove or reduce interruptions or distractions that degrade performance.	Examples of interruption or distraction include: • Frequent telephone calls • Excessive meetings • Excessive administrative tasks • Work that could be better performed by others • Excessive socializing, including social media, texting, etc.
Remove or reduce physical factors that degrade performance.	Examples of what could lessen the effects of physical work environment factors include: • Providing resources that reduce the effect of the problem, e.g., fans or heaters to correct inadequate temperature control • Communicating intentions that will remove the problem, such as working from home or planned additional office space Allow people to make reasonable adjustments that reduce the effect of problems specific to them.
Monitor progress of work environment issue resolution.	Analyze the results after the initial corrections and decide whether further correction to work environment issues is needed. If it is not possible to effectively eliminate the effect of a physical factor, pursue alternative mitigation strategies and solutions. If the initial correction to physical factors did not address the problem, review, and identify potential reasons and alternative solutions to address them.
Resolve interpersonal problems that degrade work relationships.	Examples of ways to resolve interpersonal problems include: • Improving interpersonal communication skills by mentoring or formal training • Advising or counseling one or more individuals • Using an ombudsman, arbitrator, or facilitator • Reassigning one or more individuals • Taking disciplinary action • Agreeing to disagree

Example Work Products

Example Work Products	Further Explanation
Corrections needed to the work environment	

MC 3.4

Required Practice Information

Practice Statement

Manage and resolve issues with affected stakeholders.

Value

Resolves issues early, increasing the likelihood of meeting objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Identify, communicate, and resolve issues with affected stakeholders. When notified and engaged early in the process, stakeholders can more effectively address their responsibilities to ensure they stay in sync with objectives and plans.

Examples of issues include:

- Incomplete requirements
- Design defects
- Late critical dependencies and commitments
- Solution problems
- Unavailable critical resources

Example Activities

Example Activities	Further Explanation
Identify and record issues.	Issues are commonly identified in meetings and cross-functional coordination events.
Communicate issues to affected stakeholders.	
Resolve issues with affected stakeholders.	
Escalate the issues not resolvable with affected stakeholders to the responsible managers.	
Track issues to closure.	

Example Activities	Further Explanation
Communicate the status and resolution of issues with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Recorded issues	Include the: • Issue statement • Person responsible • Due date • Resolution • Status

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Monitoring is particularly important in a complex environment. The involvement of owners, acquirers, and customers of other systems is crucial to the success of complex integrated solutions.

Monitoring is also important in an environment where a supplier is using an agile methodology in solution development activities. In such an environment, the sustained involvement of end users or their proxies can be crucial to developing and validating one or more elements of the overall provided capability.